LOW-VOLTAGE BENCH REVISION — 5V logic, 5–24VDC / 1A resistive test load only

BENCH RELAY ONLY: 5–24VDC, <=1A resistive load. No mains.

J5: contact loop only; external current-limited supply and load wired in series.

F2
Fuse (rating TBD)

C9
10uF

C10
100nF

U4
ESP32-S3-WROOM-1-N16R8

R7
4.7k

R8
4.7k

J1
SCT-013-000 ONLY

R23
18 ohm 1% 0.25W

U1
ADS1115IDGS

R13
10k

C11
1uF

SW3
RESET

R3
10k 1%

R4
10k 1%

C2
10uF

D9
BAT54S

R14
10k

SW2
BOOT

R1
1k 1%

R2
1k 1%

C1
220nF

D10
BAT54S

D1
1N4007

Q1
2N3400A

R6
100

R28
100k

J5
TEST_LOAD

J3
THERMAL_I2C

R12
330

BZ1
Buzzer (3V3 type TBD)

D5
POWER

R15
100

Q2
2N3400A

R27
100k

R26
100k

C19
10uF 10V X7R

C20
100nF 10V X7R

C3
100nF

C4
1uF

R9
1k

D2
ADC POWER

R16
330

R17
330

R18
330

D6
SMLVN6RGB1U1

AC CURRENT INPUT — CT only; never connect a voltage source to J1

SCT-013-000: 100A : 50mA assumed from project sensor choice; verify actual sensor.

18-ohm burden: 0.9Vrms / 1.27Vpk at nominal 100A; CT_BIAS = VCC/2.

ADC: differential AIN0–AIN1, +/-2.048V PGA, 860SPS, 220nF differential filter.

Calibrate 50/60Hz RMS and validate clamp behavior; this CT cannot measure DC.

GPIO35–37 reserved for octal PSRAM; unused pins intentionally NC

SDA moved from strapping GPIO3 to GPIO10; UART uses GPIO43/44

USB: self-powered 3.3V; VBUS sense only; UART without hardware flow control

U5
CP2102N-A02-QGFN28

C15
100nF

C16
4.7uF

C17
4.7uF

C18
100nF

R21
1k

R19
22.1k

R20
47.5k

D8
USB BRIDGE POWER

R22
330

J4
USB_C_Receptacle_USB2.0_14P

D7
SP0503BAHTG

R24
5.1k

R25
5.1k

BOOT: hold SW2, press/release RESET SW3, then release SW2 to enter download mode

Bench validation required: fuse/buzzer selection, regulator thermal tests and CT calibration

REGULATED 5V LOGIC INPUT — no raw battery / 12V input

J2
REGULATED 5V INPUT

F1
PTC (rating TBD)

U3
LDL1117S33R

C5
4.7uF 10V X7R

C6
100nF 10V X7R

C7
10uF 10V X7R

C8
100nF 10V X7R

Output capacitor must retain >=4.7uF under DC bias. LDO dissipation = (5–3.3)*I.

5–24VDC / 1A resistive load target. Bench validation required. No mains.

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Title: SmartDB v2 — low-voltage bench revision

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